

Curvature elasticity and red blood cell shapes

These study notes develop the calculation from a curvature energy to local force balance, then to an axisymmetric profile and a stability test. The steps include the geometric variations, integration by parts, exact substitutions, pole asymptotics, reproducible quadratures, and an explicit spherical Hessian. The logarithmic biconcave profile is treated as an interior analytic solution with singular poles; its visual resemblance to a red cell is distinguished from a smooth constrained equilibrium.

1 Curvature energy and the shape equation

The calculation starts from

$$F = \int dA \left[\frac{\kappa}{2}(2H - C_0)^2 + \bar{\kappa}K + \sigma \right] + \Delta p \int dV. \quad (1)$$

The normal points outward; H is positive on an outward-oriented sphere; Δp means exterior minus interior pressure. The sign of C_0 is fixed by $(2H - C_0)^2$. All moduli and C_0 are spatially constant. The surface is closed and smooth unless a singular analytic profile is explicitly being examined.

The total area and enclosed volume are $A = \int dA$ and $V = \int dV$. Their prescribed values are A_0 and V_0 . The notation $F, \kappa, \bar{\kappa}, H, K, C_0, \sigma, \Delta p$ follows the energy above; additional geometric symbols are defined where they first enter the calculation.

At prescribed area and volume, the last two terms are Lagrange-multiplier terms. Subtracting $\sigma A_0 + \Delta p V_0$ would make the constraints explicit but would not change a shape variation. Hold the multipliers fixed while taking that variation; solve for their values together with the shape and the two constraints. This is different from specifying tension and pressure externally and allowing area and volume to vary.

Units. Both κ and $\bar{\kappa}$ have units of energy; H and C_0 have units of inverse length; K has units of inverse length squared. σ is energy per area, and Δp is energy per volume. Thus every term in Eq. (1) is an energy. The eventual shape equation has units of energy per volume, equivalently force per area.

Expanding the square makes the role of spontaneous curvature explicit:

$$\frac{\kappa}{2}(2H - C_0)^2 = 2\kappa H^2 - 2\kappa C_0 H + \frac{\kappa C_0^2}{2}. \quad (2)$$

The last term contributes $(\kappa C_0^2/2) \int dA$; the middle term depends on the integrated mean curvature and cannot in general be absorbed into tension. Reversing the normal reverses H and C_0 together and leaves K unchanged.

For a smooth closed surface of spherical topology,

$$\int K dA = 4\pi, \quad \bar{\kappa} \int K dA = 4\pi \bar{\kappa}, \quad \delta \int K dA = 0. \quad (3)$$

This is why a constant Gaussian modulus disappears from local force balance. It does not justify dropping edge terms on a surface with a boundary, nor does it justify ignoring singular poles when integrating by parts.

The geometric variations, with the signs exposed. Let $\mathbf{X}$ denote a point of the reference surface and $\mathbf{n}$ its outward unit normal. A normal deformation is

$$\mathbf{X}_\varepsilon = \mathbf{X} + \varepsilon u \mathbf{n}, \quad \delta = \left. \frac{d}{d\varepsilon} \right|_{\varepsilon=0}.$$

Here ε is dimensionless and u is a signed scalar field with units of length. The principal curvatures are k_1, k_2 , so $2H = k_1 + k_2$ and $K = k_1 k_2$. Three first-order identities are needed:

$$\delta(dA) = 2H u dA, \quad (4)$$

$$\delta \int dV = \int u dA, \quad (5)$$

$$\delta(2H) = -\nabla^2 u - (4H^2 - 2K) u. \quad (6)$$

The Laplacian and gradient are intrinsic surface operators. Here is a geometric way to check each ingredient, rather than memorize a sign convention.

Area and swept volume. At a point choose the two principal directions. For spatially constant u , the parallel offset multiplies the two length elements by $1 + \varepsilon k_1 u$ and $1 + \varepsilon k_2 u$. Multiplying them gives

$$\frac{dA_\varepsilon}{dA} = (1 + \varepsilon k_1 u)(1 + \varepsilon k_2 u) = 1 + 2\varepsilon H u + \varepsilon^2 K u^2. \quad (7)$$

The coefficient of ε gives Eq. (4). A patch sweeps out signed volume $\varepsilon u dA$ to first order, giving Eq. (5).

Curvature change from an offset and from spatial variation. Under the same constant parallel offset, either principal curvature becomes

$$k_i(\varepsilon) = \frac{k_i}{1 + \varepsilon k_i u} = k_i - \varepsilon k_i^2 u + \mathcal{O}(\varepsilon^2), \quad i = 1, 2. \quad (8)$$

The sum of their squares is

$$k_1^2 + k_2^2 = (k_1 + k_2)^2 - 2k_1 k_2 = 4H^2 - 2K. \quad (9)$$

This accounts for the second term in Eq. (6). If the offset varies along the surface, the normal also tilts: its tangential change is minus the surface gradient of the displacement. Differentiating that tilt and taking the tangential trace adds $-\nabla^2 u$. This gives the full identity.

Sphere check. For a sphere of radius R , $K = H^2$ and a constant displacement changes $H = 1/R$ by $-H^2 u$. Equation (6) therefore gives $\delta(2H) = -2H^2 u$, exactly as differentiating $2/R$ does. The area increases under the same outward displacement.

Deriving the shape equation term by term. Apply the product rule to the bending integral, varying both curvature and area:

$$\delta \int \frac{\kappa}{2} (2H - C_0)^2 dA = \int \kappa (2H - C_0) \delta(2H) dA + \int \frac{\kappa}{2} (2H - C_0)^2 \delta(dA). \quad (10)$$

Substitute the geometric identities before doing any algebra:

$$\begin{aligned} \delta F = \int dA \Big\{ & -\kappa(2H - C_0) \nabla^2 u - \kappa(2H - C_0)(4H^2 - 2K)u \\ & + \kappa H(2H - C_0)^2 u + 2\sigma H u + \Delta p u \Big\}. \end{aligned} \quad (11)$$

The area variation is essential: omitting its bending contribution changes the nonlinear curvature terms.

For a smooth closed surface, integration by parts twice gives

$$\begin{aligned}\int (2H - C_0) \nabla^2 u \, dA &= - \int \nabla(2H - C_0) \cdot \nabla u \, dA = \int u \nabla^2(2H - C_0) \, dA \\ &= 2 \int u \nabla^2 H \, dA.\end{aligned}\tag{12}$$

The last equality uses constant C_0 . The algebraic bending terms combine as

$$\begin{aligned}& -\kappa(2H - C_0)(4H^2 - 2K) + \kappa H(2H - C_0)^2 \\ &= -\kappa(2H - C_0)[4H^2 - 2K - H(2H - C_0)] = -\kappa(2H - C_0)(2H^2 - 2K + C_0H).\end{aligned}\tag{13}$$

Consequently

$$\delta F = \int dA u \left[\Delta p + 2\sigma H - \kappa(2H - C_0)(2H^2 - 2K + C_0H) - 2\kappa \nabla^2 H \right].\tag{14}$$

Arbitrariness of the normal displacement yields

$$\boxed{\Delta p + 2\sigma H - \kappa(2H - C_0)(2H^2 - 2K + C_0H) - 2\kappa \nabla^2 H = 0.}\tag{15}$$

This is the closed-membrane force balance in the conventions of this note [1]. Curvature contains second derivatives of the shape, so $\nabla^2 H$ makes the equation fourth order in the position of the surface.

Exact checks: sphere and cylinder.

Sphere, including nonzero spontaneous curvature. Let R be the sphere radius. For a sphere, $K = H^2$, $\nabla^2 H = 0$, $\int dA = 4\pi/H^2$, and $\int dV = 4\pi/(3H^3)$. Substitution into the full energy gives

$$F = 2\pi\kappa \left(2 - \frac{C_0}{H} \right)^2 + 4\pi\bar{\kappa} + \frac{4\pi\sigma}{H^2} + \frac{4\pi\Delta p}{3H^3}.\tag{16}$$

In terms of the radius, the bending part is

$$2\pi\kappa(2 - C_0 R)^2 = 8\pi\kappa - 8\pi\kappa C_0 R + 2\pi\kappa C_0^2 R^2.\tag{17}$$

Differentiate the entire energy with respect to radius, keeping $\sigma, \Delta p, C_0$ fixed:

$$\frac{dF}{dR} = -4\pi\kappa C_0(2 - C_0 R) + 8\pi\sigma R + 4\pi\Delta p R^2.\tag{18}$$

Dividing by $4\pi R^2$ and replacing inverse radius by H gives

$$\Delta p + 2\sigma H - \kappa C_0 H(2H - C_0) = 0.\tag{19}$$

Equation (15) gives the identical result since $2H^2 - 2K + C_0H = C_0H$ on a sphere. In particular, when $C_0 = 0$,

$$\Delta p = -2\sigma H, \quad -\Delta p = \frac{2\sigma}{R}.\tag{20}$$

The bending energy is then $8\pi\kappa$, independent of size. This checks both the pressure convention and the factors of two.

Cylinder as a local test. For a cylindrical patch of radius R , $2H = 1/R$, $K = 0$, and $\nabla^2 H = 0$. For $C_0 = 0$, Eq. (15) becomes

$$\Delta p + \frac{\sigma}{R} - \frac{\kappa}{2R^3} = 0. \quad (21)$$

The energy per unit axial length, excluding end caps, is

$$\frac{\pi\kappa}{R} + 2\pi\sigma R + \pi\Delta p R^2. \quad (22)$$

Its radius derivative divided by $2\pi R$ gives the same equation. A cylinder is not a closed red-cell shape; this is a local algebraic test.

Area, volume, and a useful scaling identity. Under a uniform dilation $\mathbf{X}_\varepsilon = (1 + \varepsilon)\mathbf{X}$, the first-order changes are

$$\delta(dA) = 2dA, \quad \delta(dV) = 3dV, \quad \delta H = -H, \quad \delta K = -2K. \quad (23)$$

Use the expanded energy in Eq. (2). The integral of H^2 is scale invariant, the integral of H scales as length, and the area scales as length squared. Stationarity under this one particular variation therefore requires

$$-2\kappa C_0 \int H dA + (\kappa C_0^2 + 2\sigma) \int dA + 3\Delta p \int dV = 0. \quad (24)$$

For $C_0 = 0$ this reduces to $2\sigma \int dA + 3\Delta p \int dV = 0$. Inserting the sphere area and volume reproduces $\Delta p + 2\sigma H = 0$. This is a useful independent check on a numerical solution, but one scalar identity is not a replacement for local force balance. Its derivation assumes that the entire surface is an admissible stationary surface; a punctured solution with singular poles need not satisfy it.

The area and volume cannot be chosen independently of geometric feasibility. The isoperimetric inequality gives

$$6\sqrt{\pi} \int dV \leq \left(\int dA \right)^{3/2}. \quad (25)$$

Their dimensionless combination

$$\frac{6\sqrt{\pi} \int dV}{(\int dA)^{3/2}} \leq 1 \quad (26)$$

equals one only for a sphere. It measures volume relative to the largest volume available at that area. Deflation makes nonspherical shapes possible; it does not by itself select a unique biconcave minimum.

Useful dimensionless combinations for comparing sizes are

$$C_0 \sqrt{\frac{\int dA}{4\pi}}, \quad \frac{\sigma}{\kappa} \frac{\int dA}{4\pi}, \quad \frac{\Delta p}{\kappa} \left(\frac{\int dA}{4\pi} \right)^{3/2}. \quad (27)$$

Each is dimensionless. A claimed energy comparison at prescribed area and volume must use the same values of those constraints and the same material parameters. Comparing different members of an analytic family without that check can compare physically different problems.

2 Axisymmetric geometry and boundary conditions

Use cylindrical radius r , height z , meridional arclength s , and azimuth ϕ . Let ψ be the tangent angle, measured clockwise from the outward radial direction. Follow the upper meridian from the axis toward the rim; on this branch $\cos \psi > 0$. Thus

$$\frac{dr}{ds} = \cos \psi, \quad \frac{dz}{ds} = -\sin \psi, \quad (28)$$

and dividing gives

$$\frac{dz}{dr} = -\tan \psi = -\frac{\sin \psi}{\sqrt{1 - \sin^2 \psi}}. \quad (29)$$

The meridional principal curvature is the tangent turning rate per arclength, and the azimuthal principal curvature comes from turning around a circle:

$$k_1 = \frac{d\psi}{ds} = \frac{d \sin \psi}{dr}, \quad k_2 = \frac{\sin \psi}{r}. \quad (30)$$

With k_1 meridional and k_2 azimuthal, their sum and product give

$$2H = \frac{d \sin \psi}{dr} + \frac{\sin \psi}{r}, \quad (31)$$

$$K = \frac{\sin \psi}{r} \frac{d \sin \psi}{dr}. \quad (32)$$

For a sphere, $\sin \psi = r/R$; both expressions immediately give the positive sphere curvatures.

A ring has circumference $2\pi r$ and meridional width $ds = dr/\cos \psi$, so

$$dA = \frac{2\pi r}{\cos \psi} dr. \quad (33)$$

The surface divergence of a radial flux is its rate of change per ring area. Apply this to the intrinsic gradient of H to obtain

$$\nabla^2 H = \frac{\cos \psi}{r} \frac{d}{dr} \left[r \cos \psi \frac{dH}{dr} \right] = [1 - \sin^2 \psi] \frac{d^2 H}{dr^2} + \left[\frac{1 - \sin^2 \psi}{r} - \sin \psi \frac{d \sin \psi}{dr} \right] \frac{dH}{dr}. \quad (34)$$

The factors of cosine distinguish this from a planar radial Laplacian.

An arclength formulation and regular boundary conditions. Radius ceases to be a convenient independent coordinate at a vertical tangent. Arclength remains regular. Combining Eqs. (28) and (31) gives

$$\frac{d\psi}{ds} = 2H - \frac{\sin \psi}{r}, \quad K = \frac{\sin \psi}{r} \left(2H - \frac{\sin \psi}{r} \right). \quad (35)$$

The Laplacian becomes

$$\nabla^2 H = \frac{d^2 H}{ds^2} + \frac{\cos \psi}{r} \frac{dH}{ds}. \quad (36)$$

Hence the differential equation to integrate is

$$\frac{d^2 H}{ds^2} = -\frac{\cos \psi}{r} \frac{dH}{ds} + \frac{\Delta p}{2\kappa} + \frac{\sigma H}{\kappa} - \frac{1}{2}(2H - C_0)(2H^2 - 2K + C_0 H). \quad (37)$$

Together with the two kinematic equations and the angle equation, this is an explicit system for the meridian. H is retained as the curvature already in the energy, rather than given a new auxiliary name.

Pole regularity. At a smooth pole, radius and sine of the angle vanish, but their ratio is finite:

$$\lim_{r \rightarrow 0} \frac{\sin \psi}{r} = H_0, \quad \left. \frac{dH}{ds} \right|_{s=0} = 0. \quad (38)$$

Set $s = 0$ at the pole and write $H_0 = H(0)$ and $K_0 = K(0)$. The two principal curvatures agree at the axis. Locally,

$$\sin \psi = H_0 r + \mathcal{O}(r^3), \quad K_0 = H_0^2. \quad (39)$$

Because H has an even expansion in arclength at a smooth pole, the apparent singular term in the Laplacian has the same limit as $d^2 H/ds^2$. Taking the limit of Eq. (37) therefore gives

$$\left. \frac{d^2 H}{ds^2} \right|_{s=0} = \frac{\Delta p}{4\kappa} + \frac{\sigma H_0}{2\kappa} - \frac{1}{4} C_0 H_0 (2H_0 - C_0). \quad (40)$$

For reflection symmetry at the outer equator, impose height zero, angle $\pi/2$, and $dH/ds = 0$. These are boundary conditions for that symmetry class; they are not conditions on every possible shape.

3 The logarithmic solution and its poles

One analytic family at $\sigma = \Delta p = 0$ has the following meridional angle [2, 3]:

$$\sin \psi = C_0 r \log \left(\frac{r}{r_0} \right). \quad (41)$$

The reference radius r_0 is a fixed positive length. The logarithm has a dimensionless argument. We first check the equation on a branch with positive radius and $|\sin \psi| < 1$; closure and the poles are separate questions.

Differentiate the product of radius and logarithm explicitly:

$$\frac{d \sin \psi}{dr} = C_0 \left[\log \left(\frac{r}{r_0} \right) + 1 \right], \quad \frac{d^2 \sin \psi}{dr^2} = \frac{C_0}{r}, \quad \frac{d^3 \sin \psi}{dr^3} = -\frac{C_0}{r^2}. \quad (42)$$

Equations (31)–(32) now give

$$H = C_0 \log \left(\frac{r}{r_0} \right) + \frac{C_0}{2}, \quad (43)$$

$$K = C_0^2 \log \left(\frac{r}{r_0} \right) \left[\log \left(\frac{r}{r_0} \right) + 1 \right]. \quad (44)$$

Subtracting these expressions without discarding the constant term yields

$$H^2 - K = \frac{C_0^2}{4}, \quad \frac{dH}{dr} = \frac{C_0}{r}, \quad \frac{d^2 H}{dr^2} = -\frac{C_0}{r^2}. \quad (45)$$

For the surface Laplacian, retain its divergence form:

$$\begin{aligned}\nabla^2 H &= \frac{C_0 \cos \psi}{r} \frac{d \cos \psi}{dr} = -\frac{C_0 \sin \psi}{r} \frac{d \sin \psi}{dr} \\ &= -C_0^3 \log\left(\frac{r}{r_0}\right) \left[\log\left(\frac{r}{r_0}\right) + 1 \right] = -C_0 K.\end{aligned}\quad (46)$$

The intermediate equality uses $\cos \psi d \cos \psi = -\sin \psi d \sin \psi$.

Direct substitution and the integration constant. The two algebraic factors in the shape equation are

$$2H - C_0 = 2C_0 \log\left(\frac{r}{r_0}\right), \quad 2H^2 - 2K + C_0 H = C_0^2 \left[\log\left(\frac{r}{r_0}\right) + 1 \right]. \quad (47)$$

Their product is $2C_0 K$. The two bending contributions therefore cancel:

$$-\kappa(2H - C_0)(2H^2 - 2K + C_0 H) - 2\kappa \nabla^2 H = -2\kappa C_0 K + 2\kappa C_0 K = 0. \quad (48)$$

The residual is $\Delta p + 2\sigma H$. Since H varies with radius for nonzero C_0 , constants Δp and σ make this vanish everywhere only when both are zero. These are restrictions of this particular family.

Why a reduced equation can lose the solution. Writing C for the integration constant, an axisymmetric first integral is

$$\begin{aligned}& r [1 - \sin^2 \psi] \frac{d^2 \sin \psi}{dr^2} + \frac{r \sin \psi}{2} \left[\frac{d \sin \psi}{dr} \right]^2 + [1 - \sin^2 \psi] \frac{d \sin \psi}{dr} \\ & - \frac{\sin \psi [1 - \sin^2 \psi]}{r} - \frac{r \sin \psi}{2} \left[\frac{\sin \psi}{r} - C_0 \right]^2 - \frac{\sigma r \sin \psi}{\kappa} - \frac{\Delta p r^2}{2\kappa} = C.\end{aligned}\quad (49)$$

Differentiating its left-hand side gives $-r/\kappa$ times the left-hand side of Eq. (15), with Eqs. (31)–(34) inserted. Thus a solution of the local equation makes this expression constant.

For the logarithmic solution, the first, third, and fourth terms sum to $2C_0[1 - \sin^2 \psi]$. The second and fifth terms sum to $2C_0 \sin^2 \psi$: the difference of the two squared brackets is $4C_0^2 \log(r/r_0)$. Hence

$$C = 2C_0. \quad (50)$$

At a regular pole, Eq. (38) instead makes the first integral tend to zero. Imposing a zero constant selects the regular, force-free pole condition; it excludes this singular family. One must specify the domain and pole conditions when claiming that reduced equations are equivalent.

A continuous tangent does not imply finite curvature. For nonzero C_0 , as radius tends to zero,

$$r \log\left(\frac{r}{r_0}\right) \rightarrow 0, \quad |H| \rightarrow \infty. \quad (51)$$

Thus the slope in Eq. (29) tends to zero even though the second derivatives of the shape diverge. Let $z_0 = z(0)$; integrating the leading slope gives

$$z - z_0 = -\frac{C_0 r^2}{2} \log\left(\frac{r}{r_0}\right) + \frac{C_0 r^2}{4} + \mathcal{O}\left(C_0^3 r^4 \left| \log \frac{r}{r_0} \right|^3\right). \quad (52)$$

The tangent is continuous at the pole, but the surface is not twice continuously differentiable there. This is the distinction emphasized in modern treatments of these discs [4].

The singular bending energy is nevertheless integrable. Near one pole, dA approaches $2\pi r dr$ and $2H - C_0 = 2C_0 \log(r/r_0)$. The leading bending energy in a small cap of radial extent r_c is therefore

$$4\pi\kappa C_0^2 \int_0^{r_c} r \log^2\left(\frac{r}{r_0}\right) dr. \quad (53)$$

An explicit antiderivative of the integrand is

$$\frac{r^2}{2} \left[\log^2\left(\frac{r}{r_0}\right) - \log\left(\frac{r}{r_0}\right) + \frac{1}{2} \right]. \quad (54)$$

It tends to zero at the pole. Finite total bending energy therefore does not prove that the surface is a smooth stationary point.

The boundary term survives as a circle shrinks. Across a small circle, the outward radial flux of ∇H is

$$2\pi r \cos \psi \frac{dH}{dr} = 2\pi C_0 \cos \psi \longrightarrow 2\pi C_0. \quad (55)$$

The circumference vanishes but the curvature derivative diverges as its inverse, leaving a nonzero flux. The corresponding magnitude from the $2\kappa \nabla^2 H$ term is $4\pi\kappa|C_0|$. This is the pole force residue in this convention. A full variational treatment must retain the pole contribution or specify a supporting load; the pointwise cancellation away from the poles does not establish a smooth closed force-free equilibrium.

4 Profile reconstruction and numerical checks

Let R_e be the outer rim radius. Take an upper branch with $\sin \psi(R_e) = 1$ and $z(R_e) = 0$, then reflect it in the plane $z = 0$. Integrate Eq. (29) backward from the rim, using ρ as the dummy radial variable:

$$z(r) = \int_r^{R_e} \frac{\sin \psi(\rho)}{\sqrt{1 - \sin^2 \psi(\rho)}} d\rho. \quad (56)$$

The upper height must remain positive before it meets the reflected lower branch at the rim. A negative interior height signals intersection of this reflected construction. The angle formula alone does not check this.

Doubling the upper-half area gives

$$\int dA = 4\pi \int_0^{R_e} \frac{r}{\sqrt{1 - \sin^2 \psi}} dr. \quad (57)$$

A cylindrical shell has circumference $2\pi r$, height $2z$, and radial width dr , so

$$\int dV = 4\pi \int_0^{R_e} r z dr. \quad (58)$$

Integrate by parts using $d[r^2]/dr = 2r$. The boundary term vanishes at the pole and the zero-height rim. This gives the single integral

$$\int dV = 2\pi \int_0^{R_e} \frac{r^2 \sin \psi}{\sqrt{1 - \sin^2 \psi}} dr. \quad (59)$$

The integrand can be negative near a depressed pole; it is the integrated volume of the nonintersecting closed surface that must be positive.

Similarly, the bending contribution and integrated Gaussian curvature are

$$\int \frac{\kappa}{2} (2H - C_0)^2 dA = 2\pi\kappa \int_0^{R_e} \frac{r(2H - C_0)^2}{\sqrt{1 - \sin^2 \psi}} dr, \quad (60)$$

$$\int K dA = 4\pi \int_0^{R_e} \frac{\sin \psi d \sin \psi / dr}{\sqrt{1 - \sin^2 \psi}} dr = -4\pi \left[\sqrt{1 - \sin^2 \psi} \right]_0^{R_e} = 4\pi. \quad (61)$$

The last result is a direct improper-integral calculation for the analytic profile; it does not remove the pole regularity issue.

A reproducible biconcave example. Choose $C_0 R_e = 1.8$ and $r_0 = R_e \exp[-1/1.8]$. The same logarithmic family can then be written

$$\sin \psi = \frac{r}{R_e} + 1.8 \frac{r}{R_e} \log \left(\frac{r}{R_e} \right). \quad (62)$$

It equals one at the rim. Its minimum occurs where its derivative vanishes:

$$\frac{r}{R_e} = \exp[-2.8/1.8], \quad \min \sin \psi = -1.8 \exp[-2.8/1.8] > -1. \quad (63)$$

Thus this branch has no intermediate vertical tangent. The height maximum occurs where the slope changes sign, namely at $r/R_e = \exp[-1/1.8]$. Figure 1 is obtained by the actual slope integral, not by a polynomial fit. The value 1.8 selects an illustrative branch already discussed in the analytic-family literature [3]; the quadratures below are computed directly from Eqs. (56)–(60).

To reproduce the calculation, express all lengths in units of the rim radius, evaluate the sine in Eq. (62), and insert it directly into the four quadratures. No physiological area, volume, or bending modulus is assumed. Lengths can subsequently be restored by multiplying by the chosen rim radius; area and volume acquire its square and cube, respectively.

Endpoint treatment and numerical results. The radial integrands have an integrable square-root singularity at the rim. For this example, the derivative of the sine there is $2.8/R_e$, so, close to the rim,

$$1 - \sin^2 \psi \simeq 5.6 \left(1 - \frac{r}{R_e} \right). \quad (64)$$

For quadrature, substitute

$$\frac{r}{R_e} = \sin \theta, \quad 0 \leq \theta \leq \frac{\pi}{2}. \quad (65)$$

The quadrature coordinate θ is distinct from the tangent angle $\psi(r)$. Its Jacobian contributes $R_e \cos \theta$, which cancels the rim square-root behavior. The common transformed weight has

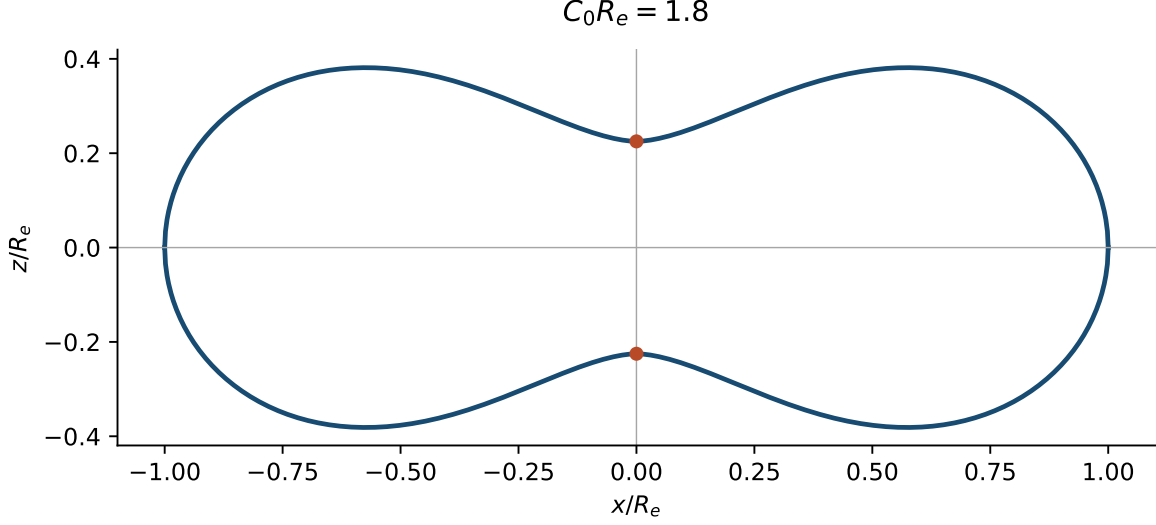


Figure 1: Meridian of the computed reflected profile. The central thickness is smaller than the maximum thickness. The orange pole markers identify points with divergent curvature, despite the continuous tangent. The horizontal coordinate $x = \pm r$ covers both sides of the meridional cross-section.

the limit

$$\frac{\cos \theta}{\sqrt{1 - \sin^2 \psi(R_e \sin \theta)}} \xrightarrow{\theta \rightarrow \pi/2} \frac{1}{\sqrt{2.8}}. \quad (66)$$

At the pole, evaluate radius times its logarithm by its zero limit, and use the explicit cap antiderivative to check the integrable logarithmic energy. Directly substituting zero into a floating-point logarithm is incorrect.

Quantity	Tolerance 10^{-7}	Tolerance 10^{-10}
z_0/R_e	0.22508429	0.22508429
$\max z/R_e$	0.38135370	0.38135370
A/R_e^2	8.87953359	8.87953359
V/R_e^3	1.93051044	1.93051044
$\frac{1}{2} \int (2H - C_0)^2 dA$	14.95016778	14.95016778
$6\sqrt{\pi} \int dV / (\int dA)^{3/2}$	0.77591314	0.77591314
$C_0 \sqrt{\int dA / (4\pi)}$	1.51308265	1.51308265
$\int K dA$	12.56637061	12.56637061

Both columns use adaptive quadrature after the endpoint substitution, with the displayed tolerance supplied as both the absolute and relative tolerance. Agreement is shown to eight decimal places; it is a convergence check, not a rigorous error bound. The Gaussian-curvature integral agrees with 4π . Here “bending energy” excludes $4\pi\bar{\kappa}$; tension and pressure terms vanish for this family. These numbers characterize the chosen geometric example and do not claim a measured red-cell energy or an energy minimum.

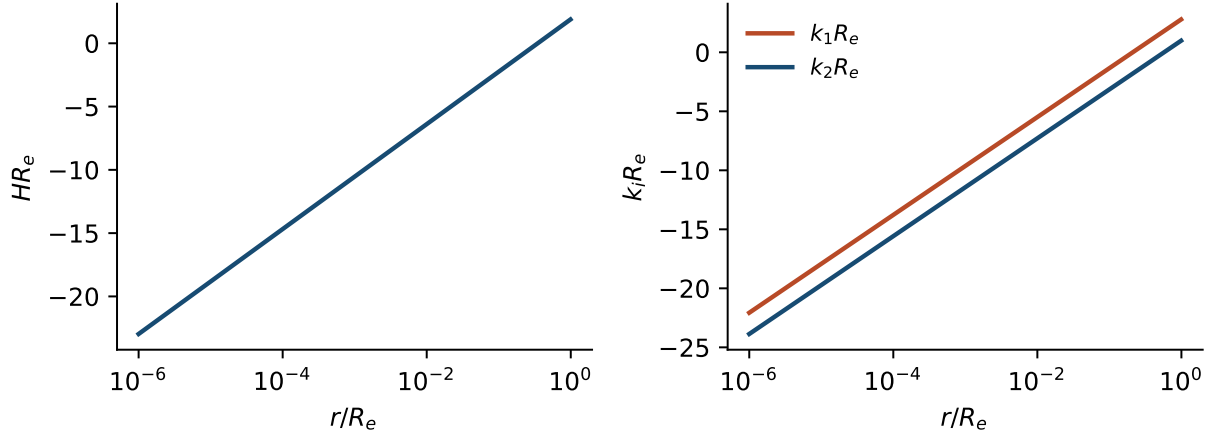


Figure 2: Curvature computed analytically from the same profile. Both principal curvatures diverge logarithmically toward the axis; their difference stays equal to C_0 . A fine-looking contour does not test this divergence.

Numerical checks of the pole qualification.

The pole singularity has two distinct numerical signatures. The bending energy of an omitted small cap tends to zero, but the first variation can retain a finite pole contribution. For one cap, integrate Eq. (60) only to the cap radius and divide its two-half prefactor by two. The leading asymptotic value follows from Eq. (53). The two calculations give:

r_c/R_e	Exact cap energy / κ	Leading asymptotic
10^{-2}	0.042704702	0.042646956
10^{-3}	0.00096096034	0.00096092873
10^{-4}	1.7112559×10^{-5}	1.7112549×10^{-5}

The same quadrature gives $\int H dA/R_e = 10.82007285$ and a dimensionless scaling residual of -10.18257345 , using $C_0 R_e = 1.8$.

Shrinking the cap thus makes very little difference to the plotted contour or the total energy, while its curvature keeps growing. Conversely, applying the smooth-surface scaling test to the entire singular example leaves

$$\frac{-2\kappa C_0 \int H dA + \kappa C_0^2 \int dA}{\kappa} \neq 0. \quad (67)$$

The nonzero value reported above is consistent with the pole residue and shows why an interior solution need not be stationary under a global rescaling. It is not a numerical failure of the pointwise shape-equation cancellation. This separates finite energy, local force balance away from the poles, and stationarity of the complete closed surface.

5 Constrained stability and membrane mechanics

Stationarity gives $\delta F = 0$. Stability asks for the sign of the quadratic change along admissible deformations. For a pure normal graph over the original surface, the area and volume expansions give

$$\delta^2 \int dA = \int [|\nabla u|^2 + 2Ku^2] dA, \quad (68)$$

$$\delta^2 \int dV = \int 2Hu^2 dA. \quad (69)$$

For area, the quadratic area-ratio term is $Ku^2 + \frac{1}{2}|\nabla u|^2$: the first contribution is the product of the two principal-direction stretches, and the second is the extra area from tilting the graph. The second variation is twice this quadratic coefficient. For volume, integrating the parallel-area factor in Eq. (7) through the swept thickness gives the quadratic coefficient Hu^2 .

First-order area and volume conservation require

$$\int 2H u dA = 0, \quad \int u dA = 0. \quad (70)$$

These alone do not make a finite perturbation satisfy the constraints. An exact constraint-preserving path generally needs a second-order normal correction v :

$$\mathbf{X}_\varepsilon = \mathbf{X} + \varepsilon u \mathbf{n} + \frac{1}{2} \varepsilon^2 v \mathbf{n} + \mathcal{O}(\varepsilon^3).$$

Its contributions to the second area and volume variations are, respectively, $\int 2H v dA$ and $\int v dA$. They must cancel Eqs. (68)–(69).

At a regular constrained stationary surface, the standard equivalent test is to evaluate the Hessian of the augmented energy F with multipliers held at their equilibrium values, on the tangent space defined by Eq. (70). The multiplier terms must remain in the Hessian even though area and volume are fixed in the physical problem. They account for the energy change associated with the second-order correction. The equivalence presumes that the chosen tangent direction extends to an admissible constrained path; the exactly spherical case has a special degeneracy discussed below.

Any admissible direction with $\delta^2 F < 0$ proves instability. Nonnegative second variation is a necessary local test; zero modes can require higher-order analysis. It is not a proof of a global minimum. The general second-variation theory is developed in Refs. [1, 7].

An explicit second-variation calculation on a sphere. Here take $C_0 = 0$ and a spherical stationary solution with $\Delta p = -2\sigma H$. Since H is constant and $K = H^2$,

$$\delta H = -\frac{1}{2}[\nabla^2 + 2H^2]u, \quad \delta K = 2H \delta H. \quad (71)$$

The second identity follows because both principal curvatures equal H : varying their product gives H times the variation of their sum. The algebraic bending term is $-4\kappa H(H^2 - K)$, whose variation vanishes:

$$\delta[H(H^2 - K)] = (H^2 - K)\delta H + H(2H\delta H - \delta K) = 0. \quad (72)$$

Varying the Laplacian of a constant leaves only $\nabla^2 \delta H$. Thus the variation of the left-hand side of the shape equation is

$$2\sigma \delta H - 2\kappa \nabla^2 \delta H = [\kappa \nabla^2 - \sigma][\nabla^2 + 2H^2]u. \quad (73)$$

The original force-balance residual is zero, so its multiplication by the varied area element contributes nothing to the second variation. Therefore

$$\begin{aligned} \delta^2 F &= \int u [\kappa \nabla^2 - \sigma][\nabla^2 + 2H^2]u \, dA \\ &= \int dA \left\{ \kappa [\nabla^2 u]^2 + (\sigma - 2\kappa H^2) |\nabla u|^2 - 2\sigma H^2 u^2 \right\}. \end{aligned} \quad (74)$$

The final expression follows from integrating the Laplacian terms by parts.

For a single spherical-harmonic displacement u of degree $\ell = 0, 1, 2, \dots$,

$$\nabla^2 u = -\ell(\ell + 1)H^2 u. \quad (75)$$

Inserting this eigenvalue into the factored operator yields

$$\delta^2 F = H^2(\ell - 1)(\ell + 2) \times [\kappa H^2 \ell(\ell + 1) + \sigma] \int u^2 dA. \quad (76)$$

Degree zero gives $-2\sigma H^2 \int u^2 dA$ (uniform size change); degree one gives zero (translations); degree two gives $4H^2(6\kappa H^2 + \sigma) \int u^2 dA$. For nonnegative tension all degrees at least two are positive.

What the spherical calculation can and cannot establish. The previous Hessian is directly a test in the prescribed-tension, prescribed-pressure ensemble. In that ensemble a uniform-radius variation is allowed, so its sign must also be considered. If $\sigma > 0$, the degree-zero direction is negative; dropping it without a size constraint would incorrectly declare full stability.

With both area and volume exactly fixed at their spherical values, the isoperimetric inequality permits no nearby nonspherical embedded surface. The first-order constraint conditions coincide on a sphere and hide this restriction. The second-order conditions expose it. Subtract $2H$ times the second volume constraint from the second area constraint; the unknown second-order normal correction cancels, leaving

$$\int [|\nabla u|^2 - 2H^2 u^2] \, dA = 0. \quad (77)$$

For a pure harmonic degree this is

$$H^2[\ell(\ell + 1) - 2] \int u^2 dA = 0. \quad (78)$$

After volume conservation excludes degree zero, degrees at least two have positive coefficients; only translations survive. The sphere formulas are therefore excellent checks of an operator or a pressure-controlled problem, but they do not prove the stability of a deflated biconcave membrane.

Why nonaxisymmetric perturbations must be tested. On an axisymmetric base surface, a perturbation can vary both along the meridian and around the azimuth. Its surface Laplacian includes

$$\nabla^2 u = \frac{1}{r} \frac{\partial}{\partial s} \left[r \frac{\partial u}{\partial s} \right] + \frac{1}{r^2} \frac{\partial^2 u}{\partial \phi^2}. \quad (79)$$

A dependence $\cos(2\phi)$ or $\sin(2\phi)$ contributes $-4/r^2$ times the displacement through the last term; threefold dependence contributes $-9/r^2$. Omitting the azimuthal term excludes precisely the directions needed to test elliptical and polygonal deformations. Pressure-induced polygonal transformations were studied for biconcave vesicles and liposomes [5]; those calculations should not be read as a measured instability threshold for human red blood cells.

A calculation workflow for smooth constrained shapes.

1. **Specify the problem.** Prescribe area and volume, κ, C_0 , topology, and the allowed symmetry class. Check Eq. (26). Decide whether area and volume are constraints or whether pressure and tension are externally fixed.
2. **Integrate the geometry and force balance.** Use the arclength equations with the pole expansions above. Adjust the starting curvature, the multipliers, and the arclength interval to meet the other pole or the reflection-plane conditions and both global constraints. Along the full meridian, area accumulates at $2\pi r$ per unit arclength. For an outward orientation from the upper pole to the lower pole, volume accumulates at $\pi r^2 \sin \psi$ per unit arclength.
3. **Check the computed equilibrium.** Evaluate every term in Eq. (15) on a refined grid, test constancy of Eq. (49), and verify its zero value for regular poles. Check area, volume, closure, nonintersection, pole regularity, and Eq. (24). A small area error does not imply a small local force residual.
4. **Compute the constrained second variation.** Differentiate the augmented energy or the full force residual about the smooth base shape, keeping multipliers fixed. Away from a sphere, varying $\nabla^2 H$ also varies the surface metric, and $\delta K = 2H\delta H$ is generally false. Reusing Eq. (74) on a biconcave profile therefore misses terms. Project perturbations onto the two first-order constraints and include azimuthal dependence.
5. **Verify the Hessian numerically.** For a fixed trial displacement, compare the quadratic form to

$$\frac{F(\varepsilon) - 2F(0) + F(-\varepsilon)}{\varepsilon^2} = \delta^2 F + \mathcal{O}(\varepsilon^2). \quad (80)$$

Here $F(\varepsilon) = F[\mathbf{X} + \varepsilon \mathbf{u}\mathbf{n}]$, with u held fixed. The actual quadratic energy increment is $\frac{1}{2}\varepsilon^2 \delta^2 F$. Reduce ε until the expected convergence is visible, before subtraction roundoff dominates. Repeat with spatial refinement. Include the multiplier terms, or instead re-enforce both constraints to second order before comparing the bending energies.

6. **Interpret the outcome within its domain.** A negative constrained direction proves instability of that stationary shape. Testing only one analytic family or only rotationally symmetric directions leaves other instabilities untested. Singular profiles require explicit pole boundary or load conditions before the same stability machinery can be applied.

Connection to real red-cell mechanics. The fluid-bilayer energy accounts for curvature elasticity. A red-cell membrane also has a protein skeleton that supports shear and stretching. Combining bilayer mechanics with skeleton elasticity can reproduce cup-like, biconcave, and spiculated shape sequences [6]. The detailed bilayer calculation above supplies one part of that model, not a complete constitutive description of a living cell.

For a quantitative comparison, state area, volume, leaflet asymmetry, external loading, and whether the skeleton is included. Static F supplies equilibrium and energetic stability, not deformation rates; dynamics additionally needs dissipation and transport. A useful finished calculation therefore reports the shape, its regularity and constraints, and the perturbations whose energy was actually tested.

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